

ME416

Ground Vehicle Dynamics

Kinematics,
Roll Center

Fall 2020

[1]

Suspension Design

- We have introduced an initial conceptual model (bicycle and four-wheel models) of a steady-state cornering vehicle
 - These models were very simplistic but gave us a general idea of how lateral forces are generated and the geometrical origins of steer angles, slip angles, forces and moments
 - We will revisit steady-state cornering and those models in a later lecture but, as engineers, we did recognize some aspects and limitations of the models that we would need to address
 - Static load distribution
 - Dynamic load distribution due to lateral and longitudinal forces
 - We know that the distribution of forces on a mechanism depends on the kinematics of the mechanism, i.e. ICR, c.g. and geometry
 - Therefore, to design an effective suspension we must be aware of the effects of the kinematics of various suspension parameters and design types to aid us in achieving our intended goals for the vehicle

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Suspension Design (2)

- We have two major areas of concern
 - Ride
 - Previously, we may have only thought of ride as the isolation of vehicle occupants or payload from road irregularities and other excitations
 - We can also consider ride in terms of vehicle attitude changes due to longitudinal or lateral acceleration
 - For production cars certain motions are less disturbing than others
 - Largely a result of evolutionary factors
 - What motions do you think are preferred or disliked?
 - For high performance cars the primary concern is vehicle attitude effects on performance
 - Aerodynamics
 - Knock-on effects on suspension geometry
 - Handling
 - The suspension affects handling directly as it defines
 - Geometry of the motion of the vehicle and wheels
 - Load transfer
 - Additional effects include tire wear, noise and stability

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Suspension Design (3)

- We will now focus on how the sprung and unsprung masses are connected and dictate
 - Relative motion of the sprung and unsprung mass
 - How the forces are transmitted between the sprung and unsprung mass
 - The sprung mass is the mass supported on the springs of the suspension such as
 - Chassis
 - Engine
 - Passenger/payload
 - The unsprung mass is the unsupported mass of the vehicle such as
 - Wheels
 - Tires
 - Brakes
 - Upright
- Any particular geometry must be designed to meet the vehicle's particular needs

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There is no single best geometry

Degrees of Freedom and Motion Path

- The suspension of a vehicle is designed such that the wheel follows a particular fixed path of motion relative to the sprung mass
- For any body moving in space we can define its motion by
 - Three components of linear motion
 - Three components of rotational motion
- These six components are generally referred to as Degrees of Freedom (DOF), i.e. in three dimensions a body has 6 DOF
- For a vehicle suspension we have
 - Vertical – also called Bump
 - Lateral
 - Longitudinal
 - Steer Axis
 - Wheel Spin Axis
 - Camber, inclination or overturn
- Suspensions essentially limit the freedom of or constrain the movement of the wheel in 4 to 5 DOF
 - Independent suspension: 1 DOF in the ride (vertical) direction
 - Solid axle or beam-type suspension: 2 DOF in the ride and roll directions

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Suspension Components

- For simplicity a suspension can be considered to consist of rigid links
- If the links are tension-compression only links, i.e. two force members, then to constrain the wheel path in X DOF requires X Links
- Consider:
 - How many TC links are required for an independent suspension?
 - How many TC links are required for a solid axle or beam-type suspension?
 - How many DOF does an A-arm or wishbone constrain?
 - How many DOF does a MacPherson or Chapman strut constrain?
- We will now consider some fundamental simplifications and implications of various suspension geometries

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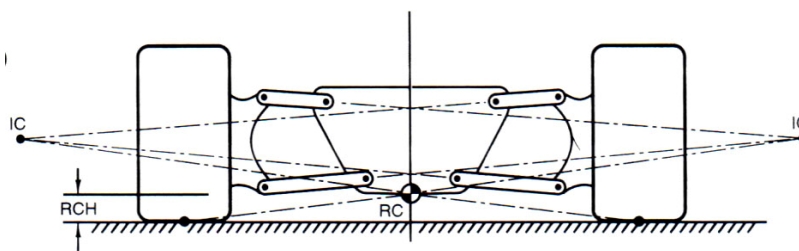
Roll Center

- The roll axis is the axis about which the lateral acceleration forces acting at the c.g. are resolved about
- The roll axis is defined by a line passing through the roll centers of the front and rear suspension
- The roll centers are kinematically defined as the Instantaneous Center of Rotation (ICR) of the front and rear suspensions
 - Recall for ICR's
 - Defined as the point at which a body has zero instantaneous velocity such that the body appears to be rotating about it at that instant
 - The ICR will vary with geometry and will move relative to the vehicle body and ground
- The roll center is the virtual point where tire road forces are transferred through the suspension arms and act upon the chassis

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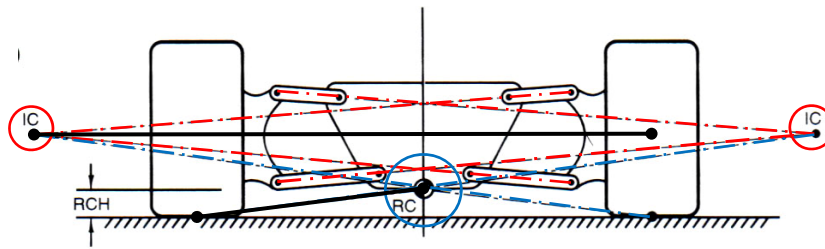
Roll Center – Double A-Arm



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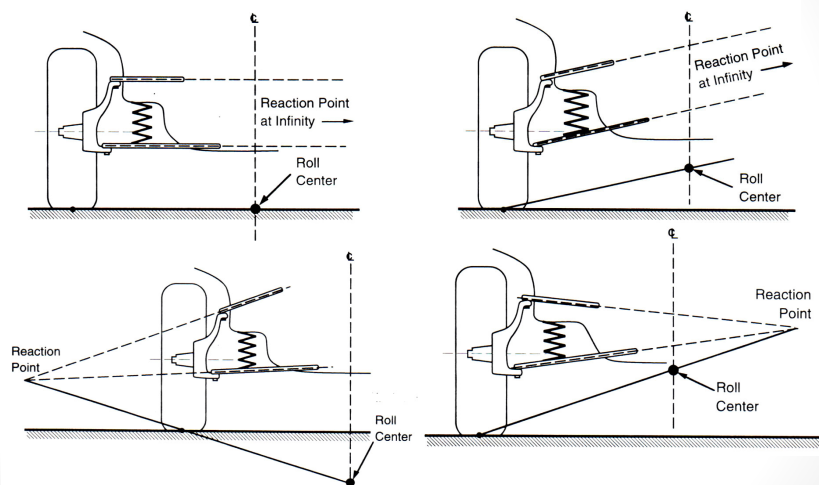
Roll Center – Double A-Arm



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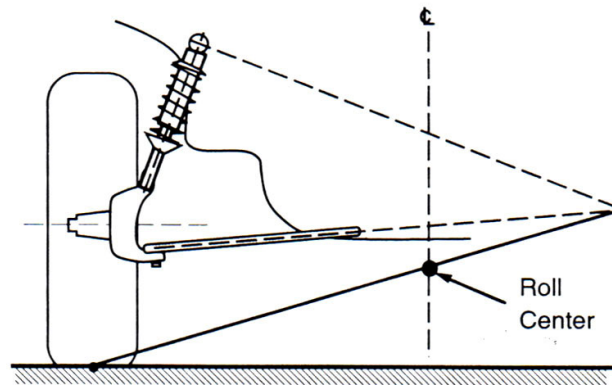
Roll Center – Double A-Arm (2)



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[10]

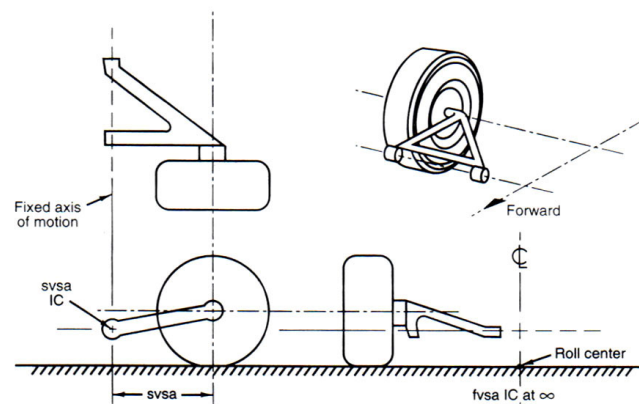
Roll Center – MacPherson/Chapman



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[11]

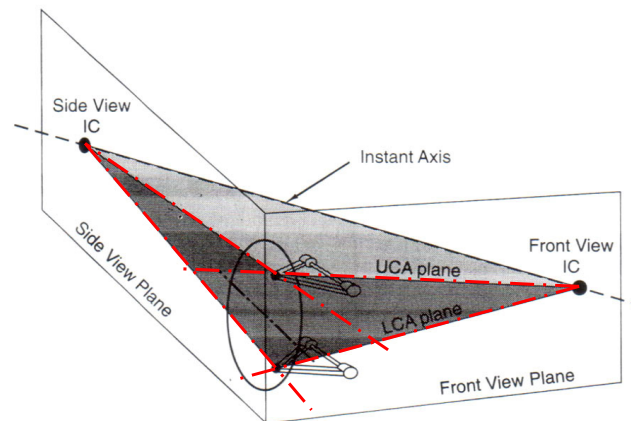
Roll Center – Trailing Arm



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[12]

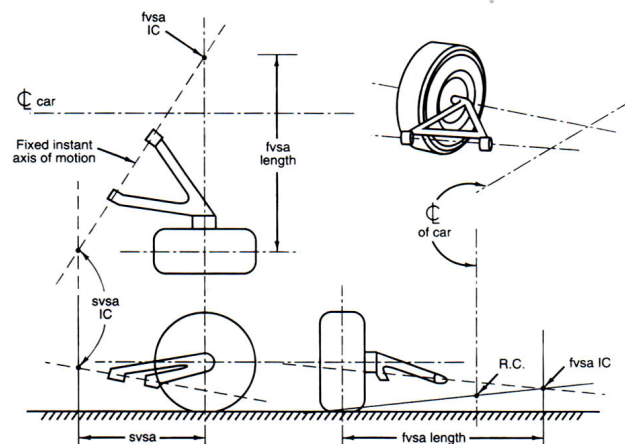
Roll Center – Inclined Pivots



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[13]

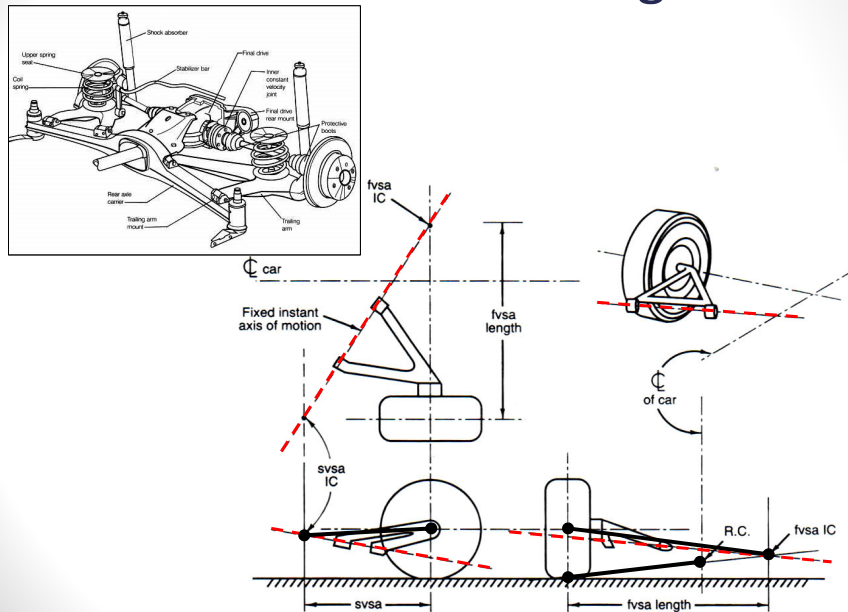
Roll Center – Semi-Trailing Arm



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[14]

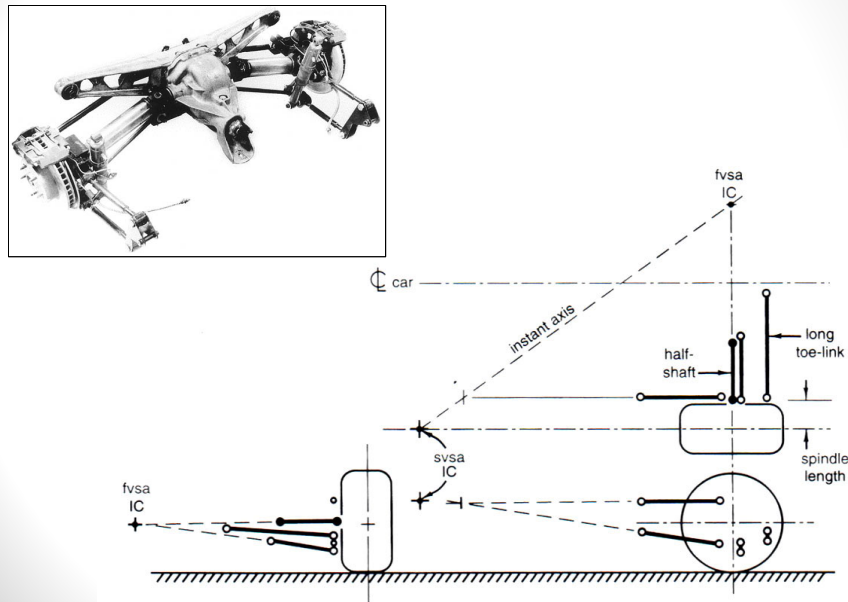
Roll Center – Semi-Trailing Arm



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[15]

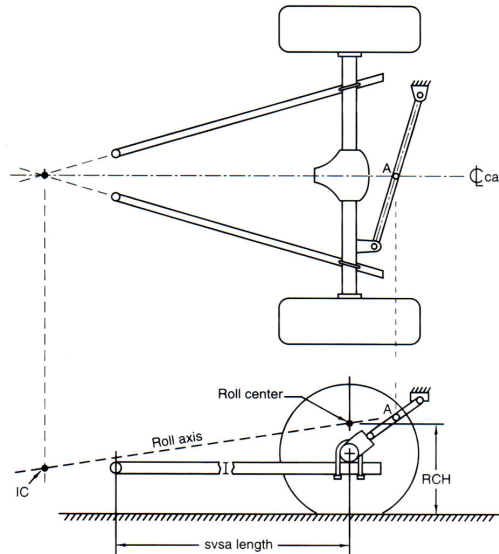
Roll Center – 5-Link Corvette Rear



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[16]

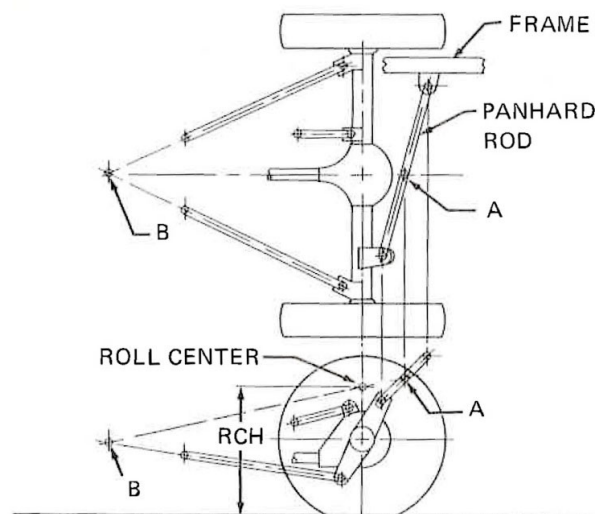
Roll Center – NASCAR Live Axle



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[17]

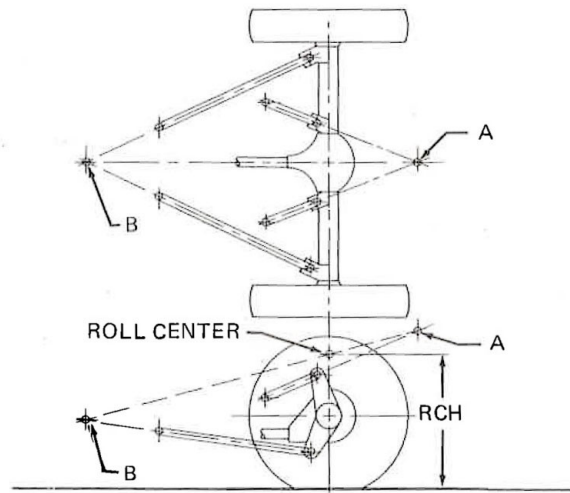
Roll Center – 3-Link, Panhard



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[18]

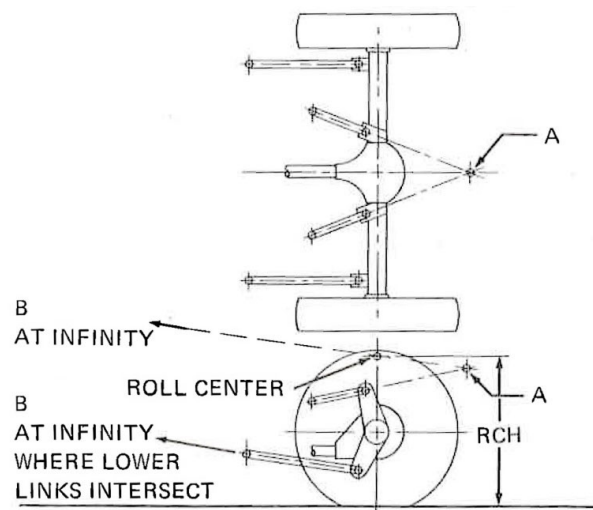
Roll Center – 4-Link, Non-Parallel



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[19]

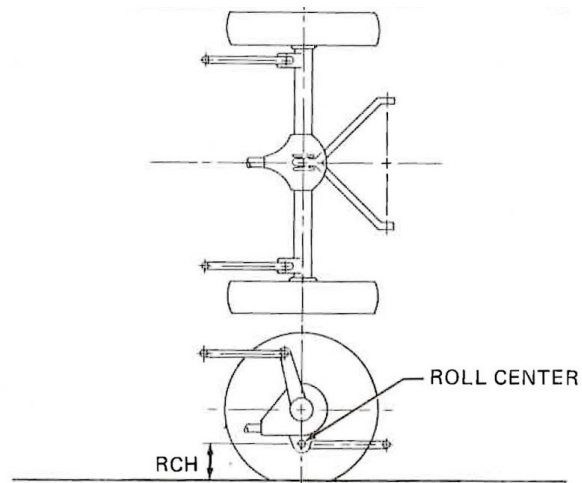
Roll Center – 4-Link, Parallel



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Roll Center – Pivot (Watts, Panhard Mod)

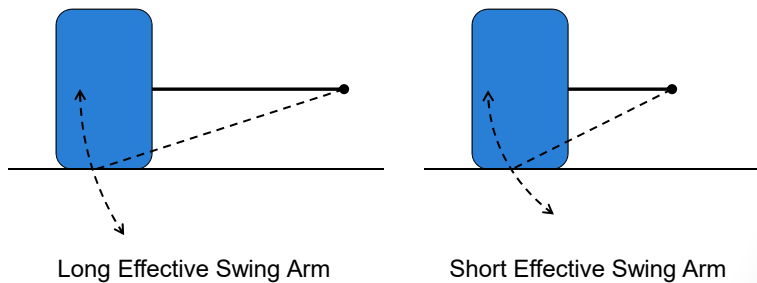


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Tire Scrub

- Consider the motion of a tire relative to the ground plane of the following two suspension geometries:



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Tire Scrub (2)

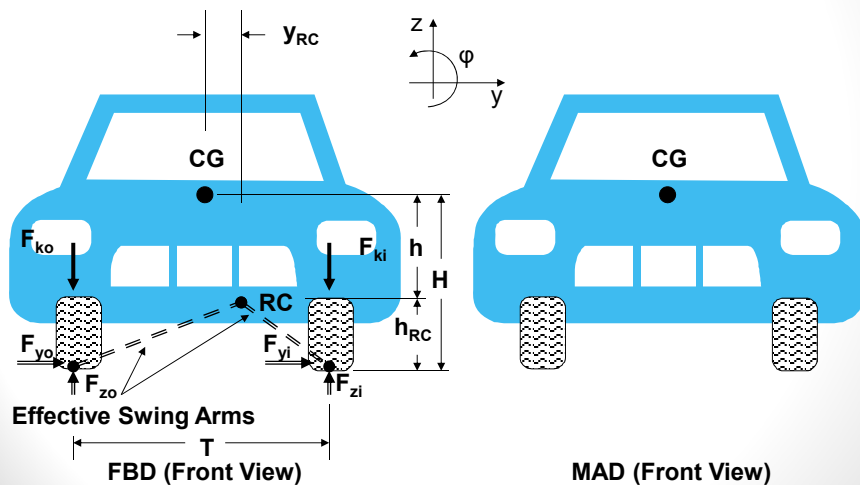
- The lateral motion of the tire across the ground is called scrub
 - Scrub is undesirable due to increased tire wear and other effects
 - Scrub is always(...) minimized in suspension design
- If we assume that we will design a suspension with minimal scrub then we can assume a simple single DOF system to represent the mechanical system of chassis, suspension and tire
 - This single DOF is the pivoting of the tires and corresponds to the determination of the suspension RC
 - For all intents and purposes we assume all tire motion is vertical

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Kinematics Analysis (lannce)

- Consider the forces acting on the tires/suspension shown:

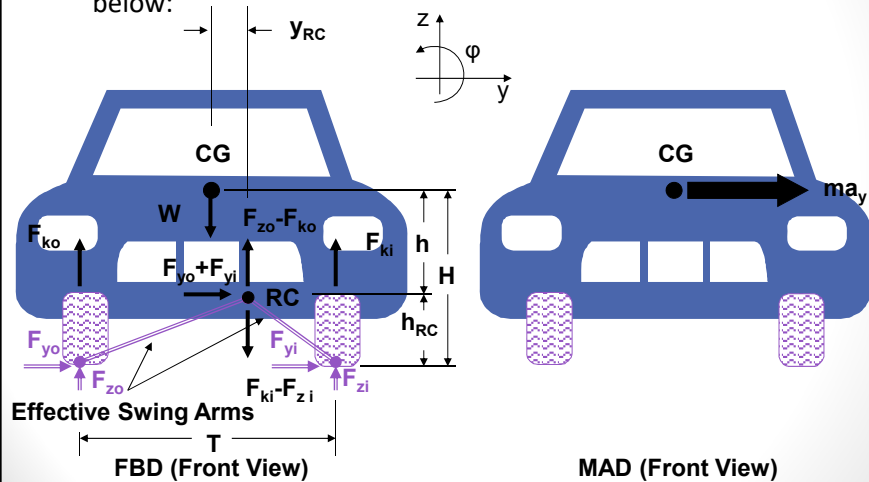


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Kinematics Analysis (lannce)

- Assume the roll-plane FBD and MAD of the sprung mass below:

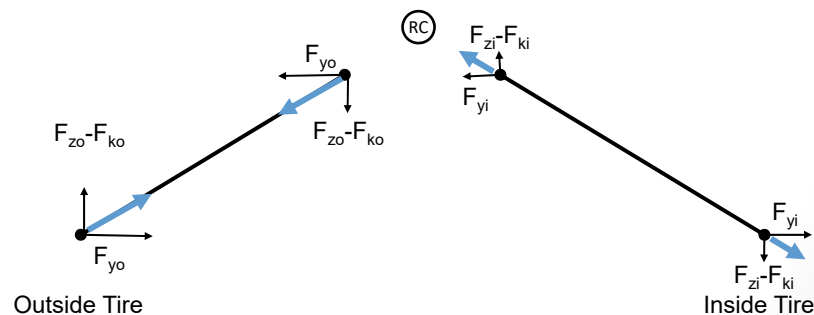


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Kinematics Analysis – Effective Swing Arms

- We can define the effective swing arms from the tire/ground contact point to the RC of the suspension



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Kinematics Analysis – Equilibrium Equations

- From the FBD and MAD the equilibrium equations are

$$\begin{aligned}\sum F_y : F_{yi} + F_{yo} &= ma_y \\ \sum F_z : F_{zo} - F_{ko} + F_{zi} - F_{ki} + F_{ko} + F_{ki} - W &= 0 \\ \sum M_{RC} : \left(\frac{T}{2} - y_{RC}\right) F_{ki} + y_{rc} W - \left(\frac{T}{2} + y_{RC}\right) F_{ko} &= -hma_y\end{aligned}$$

- The lateral forces F_{yi} and F_{yo} are statically indeterminate
 - These forces depend on the tire properties and steering geometry
 - For now we will assume F_{yi} and F_{yo} are known
- The normal tire forces F_{zi} and F_{zo}
 - Will also be developed later
 - For now we will assume F_{zi} and F_{zo} are known

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Kinematics Analysis – Spring Forces

- Simplifying the equilibrium equations yields

$$\begin{aligned}\sum F_y : F_{yi} + F_{yo} &= ma_y \\ \sum F_z : F_{zi} + F_{zo} &= W \\ \sum M_{RC} : \left(\frac{T}{2} - y_{RC}\right) F_{ki} + y_{rc} W - \left(\frac{T}{2} + y_{RC}\right) F_{ko} &= -hma_y\end{aligned}$$

- We can define the spring forces as functions of bump (ζ) and roll (ϕ)

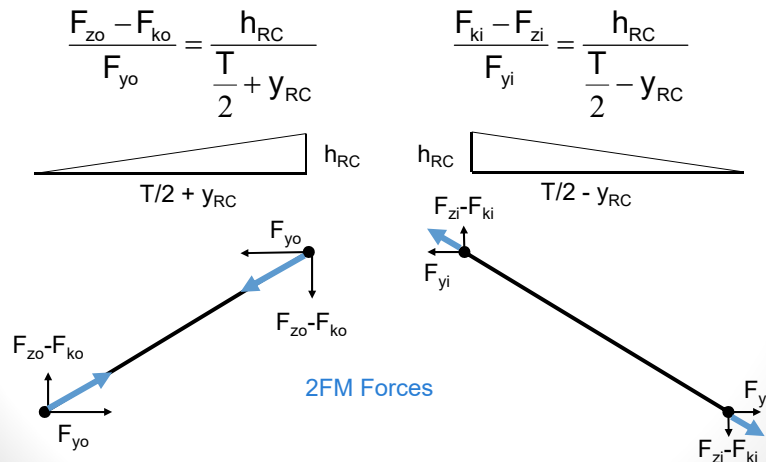
$$\begin{aligned}F_{ki} &= k_r \left(\zeta - \frac{T}{2} \phi \right) + F_{PL} \\ F_{ko} &= k_r \left(\zeta + \frac{T}{2} \phi \right) + F_{PL}\end{aligned}$$

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Kinematics Analysis – Effective Swing Arms

- The effective swing arm is a two-force member and this constraint yields the forces on the sprung mass



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Kinematics Analysis – Bump and Roll

- Solving the spring forces for bump (ζ) and roll (ϕ)

$$\zeta = -\frac{1}{2k_r}(F_{ko} + F_{ki} - 2F_{PL})$$

$$\phi = \frac{1}{Tk_r}(F_{ko} - F_{ki})$$

- Solving the swing arm constraints for F_{ko} and F_{ki}

$$F_{ko} = F_{zo} - \frac{h_{RC}F_{yo}}{\frac{T}{2} + y_{RC}}$$

$$F_{ki} = F_{zi} + \frac{h_{RC}F_{yi}}{\frac{T}{2} - y_{RC}}$$

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Kinematics Analysis – Combining

- Solving the spring forces for bump (ζ) and roll (ϕ)

$$\zeta = -\frac{1}{2k_r} \left[W + \left(\frac{F_{yi}}{\frac{T}{2} - y_{RC}} - \frac{F_{yo}}{\frac{T}{2} + y_{RC}} \right) h_{RC} - 2F_{PL} \right]$$

$$\phi = \frac{1}{Tk_r} \left[F_{zo} - F_{zi} - \left(\frac{F_{yi}}{\frac{T}{2} - y_{RC}} + \frac{F_{yo}}{\frac{T}{2} + y_{RC}} \right) h_{RC} \right]$$

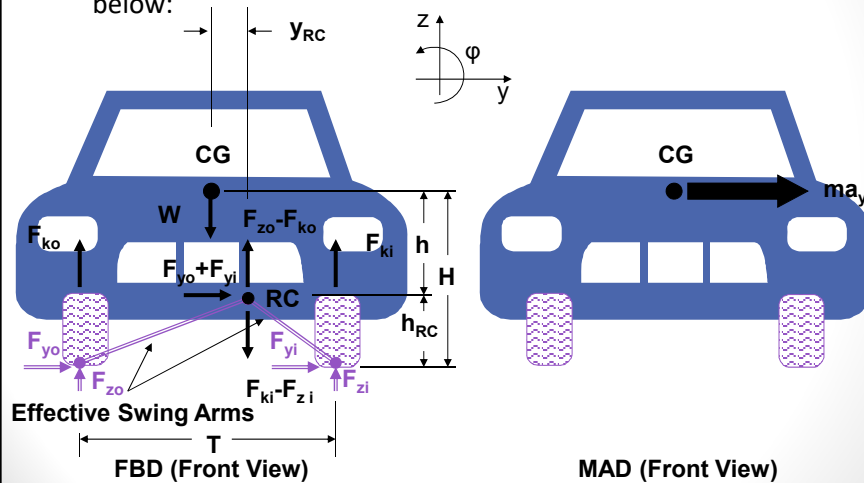
- Note the ride height and roll angle will increase dramatically as the RC approaches $y_{RC} = T/2$
 - According to Ianncce, you should keep the RC to within $\pm 5\%$ of the track width from centerline
 - In non-standard conditions you may have need for a RC outside of the vehicle track
 - Any ideas what non-standard conditions?
 - In this case, the RC must be controlled to not move significantly as well and to not approach $y_{RC} = T/2$

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Kinematics Analysis (Ianncce)

- Assume the roll-plane FBD and MAD of the sprung mass below:



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Kinematics Analysis – Simplifying

- Assuming $y_{RC}=0$, i.e. on the centerline of the vehicle

$$\zeta = -\frac{1}{k_r} (F_{y0} - F_{yi}) \frac{h_{RC}}{T}$$

$$\phi = \frac{1}{T k_r} \left[F_{z0} - F_{zi} - \frac{2 m a_y h_{RC}}{T} \right]$$

- Summing moments about the centerline at the ground yields

$$F_{z0} - F_{zi} = \frac{2}{T} (h_{RC} + h) m a_y = \frac{2}{T} H m a_y$$

- Substituting into the simplified roll approximation above

$$\phi = \frac{2}{T^2 k_r} m a_y (H - h_{RC})$$

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Kinematics Analysis – Jacking

- From the equations for bump (ζ) and roll (ϕ) below

$$\zeta = -\frac{1}{k_r} (F_{y0} - F_{yi}) \frac{h_{RC}}{T}$$

$$\phi = \frac{2}{T^2 k_r} m a_y (H - h_{RC})$$

- We note that as the RC height increases the bump increases as well
- This behavior is called jacking and is a problem in suspensions that have a high RC
 - Examples?
 - Can you say “Unsafe at Any Speed”?
- lancc recommends that the RC height be kept close to the ground
 - 5-10% of track is a good rule of thumb

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Roll Stiffness

- The roll stiffness of a suspension describes its resistance to roll
 - A rotational spring rate with units of moment per radian (or degree)
 - We can define the roll stiffness
 - For the roll moment divided by roll angle

$$k_{\phi} = \frac{(H - h_{RC})ma_y}{\phi}$$

- Substitute the previous roll relation

$$k_{\phi} = \frac{(H - h_{RC})ma_y}{\frac{2}{T^2 k_r} ma_y (H - h_{RC})}$$

$$k_{\phi} = \frac{1}{2} T^2 k_r$$

- The roll stiffness can be modified by changing
 - Spring rate
 - Track
- Note that the roll of the vehicle is greatly influenced by the c.g. height but the roll stiffness is not a function of c.g. height